



Department of Mathematics and Natural Sciences
Final Examination
Semester: Fall 2023
Course Title: Mathematics II: Integral Calculus and Differential Equations
Total marks: 30
Time: 1 hour

Instructions:

- Please write your name, ID, and section on your script.
- Give the page number (use pen) in the right corner of each page.
- Please read the question paper very carefully.
- You have to answer all the questions.

1. Evaluate $\iint_R \sqrt{16 - y^2 - x^2} \, dA$; R is the region in the first quadrant within the circle

$$x^2 + y^2 = 16. \quad [6]$$

2. Use cylindrical coordinates to evaluate, $\int_0^a \int_0^{\sqrt{a^2 - x^2}} \int_0^{a^2 - x^2 - y^2} x^2 \, dz \, dy \, dx.$ [6]

3. Determine whether the given differential equation is exact. If it is exact, then solve it. [6]

$$\left(y(1 + \frac{1}{t}) + \cos y\right)dt + (t + \log t - t \sin y)dy = 0.$$

4. Solve the given differential equation by undetermined coefficient. [6]

$$y'' - 6y' - 25y = 64e^{-x}$$

5. The population of a certain country is known to increase at a rate proportional to the number of people presently living in the country. If after two years the population has doubled, and after three years the population is 20,000, estimate the number of people initially living in the country. [6]